

Module 1 CS- Introduction

1 what is meaning of cyber security

- Cyber Security is the practice of protecting computers, networks, systems, and data from cyber attacks, hacking, viruses, and unauthorized access.

2.what are the main objectives of cyber security?

The main objectives of cyber security are:

1. Confidentiality – Protecting data from unauthorized access.
2. Integrity – Ensuring data is accurate and not modified illegally.
3. Availability – Making sure systems and data are available when needed.
4. Authentication – Verifying the identity of users.
5. Non-Repudiation – Preventing denial of actions performed online.

3.What is offensive and defensive in cyber security?

Offensive Security

Offensive security means testing systems by attacking them ethically to find weaknesses before hackers do.

Examples:

- Penetration Testing
- Ethical Hacking
- Vulnerability scanning

Defensive Security

Defensive security means protecting systems from cyber attacks.

Examples:

- Firewalls
- Antivirus
- Intrusion Detection Systems
- Data Backup

4.what is cyberspace and law

Cyberspace

Cyberspace is the virtual environment of the internet where communication, online activities, and digital data exist.

Cyber Law

Cyber law is the set of rules and regulations that deal with internet usage, cyber crimes, online transactions, and digital security.

Examples:

- IT Act 2000 in India
- Laws against hacking and online fraud

5. What is cyber welfare?

Cyber welfare refers to the safe, secure, and responsible use of the internet for the benefit of society.

It focuses on:

- Online safety
- Digital awareness
- Protection from cyber crimes
- Responsible internet usage

6.Explain the Types of Hacker

1. White Hat Hacker

Ethical hackers who help improve security legally.

2. Black Hat Hacker

Hackers who illegally steal data or damage systems.

3. Grey Hat Hacker

Hackers who may break rules without permission but usually not for harmful purposes.

4. Script Kiddies

Beginners who use ready-made hacking tools without deep knowledge.

5. Hacktivists

Hackers who attack systems for political or social causes.

6. State-Sponsored Hackers

Hackers working for governments to perform cyber espionage or attacks

7.What is the full form of SOC in cyber security

SOC stands for Security Operations Center.

It is a centralized team or facility that monitors, detects, analyzes, and responds to cyber security threats.

8.What are the Challenges of Cyber Security

Rapidly changing technology

- Increase in cyber attacks
- Lack of cyber security awareness
- Phishing and social engineering attacks
- Malware and ransomware threats
- Data privacy issues
- Shortage of skilled cyber security professionals
- Securing cloud and IoT devices
- Managing large amounts of sensitive data
- High cost of security implementation